Harry Potter Knowledge Graph

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Abstract

Goal

Translating the complex Harry Potter domain into a machine-readable RDF Knowledge Graph.

Logic

Utilizing OWL and Description Logic to support semantic modeling and reasoning..

Innovation

Combining formal logic-based reasoning with modern AI tools (LLMs) to bridge natural language questions and structured semantic queries

Requirements & Competency Questions

Ontology Goals:

- Design an OWL ontology for the Harry Potter universe.
- Represent domain knowledge as RDF triple.
- Populate and query the knowledge graph using GraphDB and SPARQL.
- Explore the use of LLMs for semi-automatic knowledge extraction

Competency Questions (CQs):

- Which students belong to Gryffindor?
- Who teaches Potions?
- Who are Harry Potter's friends?
- Who casts the "Expelliarmus" spell?
- Who is Voldemort's enemy?
- Can a student be inferred as a Wizard?

Related Work & Existing Ontologies

Existing Works:

- ✔ Pianzola, F., et al. (2025). The GOLEM Ontology for Narrative and Fiction. Humanities, 14(10), 193.
- ✔ Scotti, L., et al. (2025). Grounding the Development of Ontologies for Fictional Characters.
- ✔ Noy, N. F., & McGuinness, D. L. (2001). Ontology Development 101.
- ✔ Yani, M., & Krisnadhi, A. A. (2021). Challenges, Techniques, and Trends of Simple KGQA.
- ✔ D'Abramo, J., et al. (2025). Investigating Large Language Models for Text-to-SPARQL Generation.

Our Motivation:

- ✔ **Building a Logically Rich Ontology:** Moving beyond general models to implement formal OWL Description Logic with disjoint classes and existential restrictions.
- ✔ **Evidence-based Character Modeling:** Adopting GOLEM and Scotti et al.'s methodologies for cognitively-grounded, flexible character representation.
- ✔ **Advanced Semantic Querying:** Implementing state-of-the-art LLM-based NL-to-SPARQL architectures to bridge natural language with our Knowledge Graph.
- ✔ **Robust Evaluation Framework:** Applying standardized KGQA assessment techniques to ensure high system accuracy and reliability.

Ontology Design Rationale

Why these classes?

- ✔ Guided by established ontology engineering methodologies and upper-level ontologies for fictional narratives.
- ✔ Harry Potter provides a rich, well-defined domain with clear entity types and relationships.
- Supported by studies on:

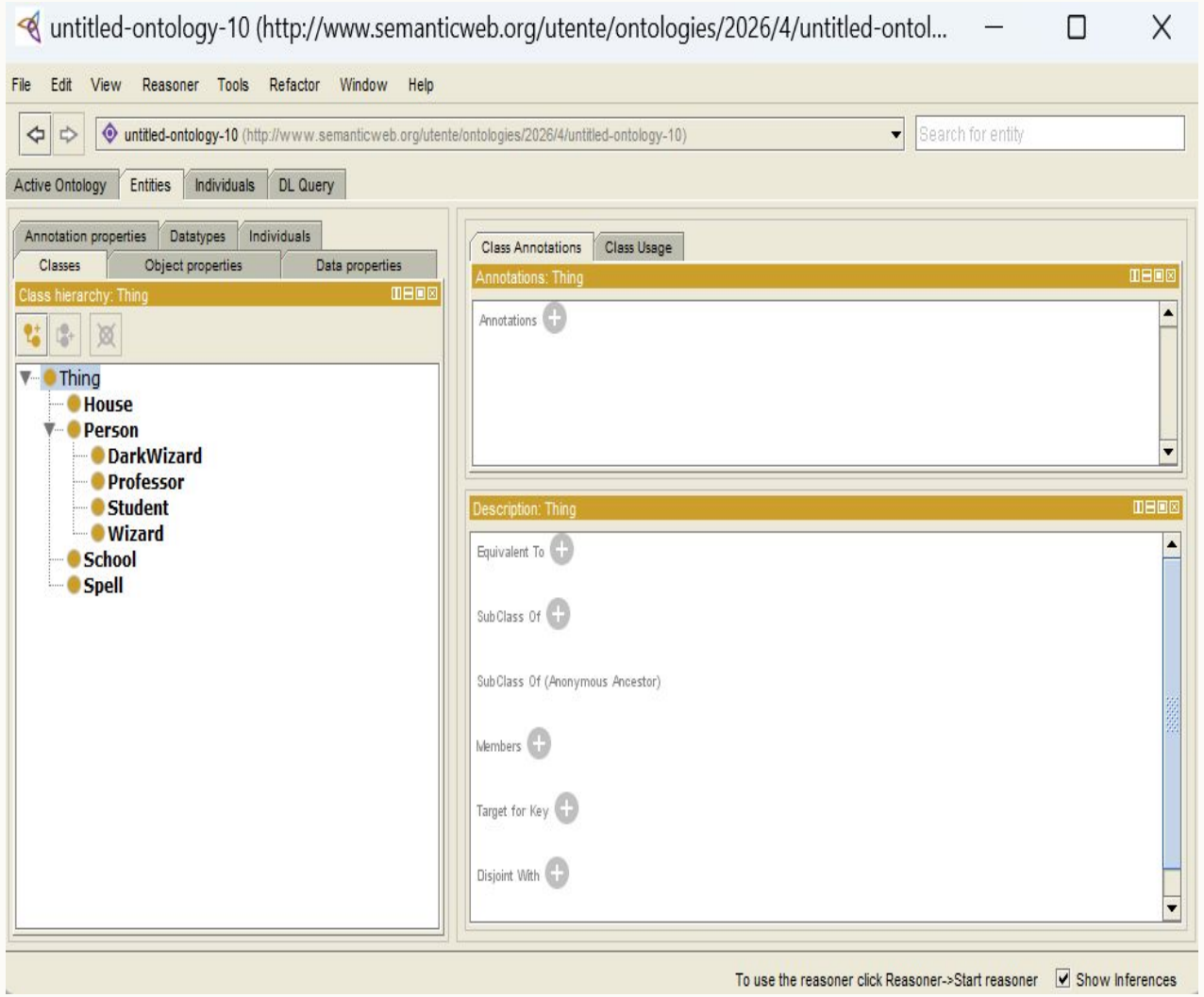
- ❑ Pianzola, F., et al. (2025). The GOLEM Ontology for Narrative and Fiction. *Humanities*, 14(10), 193.
- ❑ Scotti et al. (2025) – Grounding the Development of Ontologies for Fictional Characters.

Main Classes :

Following GOLEM's modular approach to fictional entities, we established **Person** as the foundational superclass. This provides a cognitively grounded and interoperable upper-level class for all narrative agents. GOLEM emphasizes flexible, non-hierarchically rigid modeling of fictional characters. We deliberately modeled **Student** and **Wizard** as **direct siblings** (rather than nesting Student under Wizard). This design choice maximizes the expressive power of Description Logic reasoning.

Main Design Decisions:
Professor $\perp$ Student (disjoint classes)
Student $\subseteq \exists$ belongsToHouse.House (existential restriction)
Symmetric (friendOf) and inverse (teaches / taughtBy) properties
Main Classes: Person, Wizard, Student, Professor, DarkWizard, House, School, Spell

Ontology Structure



Class Hierarchy

As explained in the previous slide, we chose a flexible hierarchy inspired by GOLEM.

In Protégé we can see:
Student, Professor, Wizard, and DarkWizard modeled as **siblings** under Person
This structure enables powerful automated reasoning (e.g., inferring Wizard status via castsSpell property)
While maintaining clear separation (e.g., Professor and Student are disjoint)

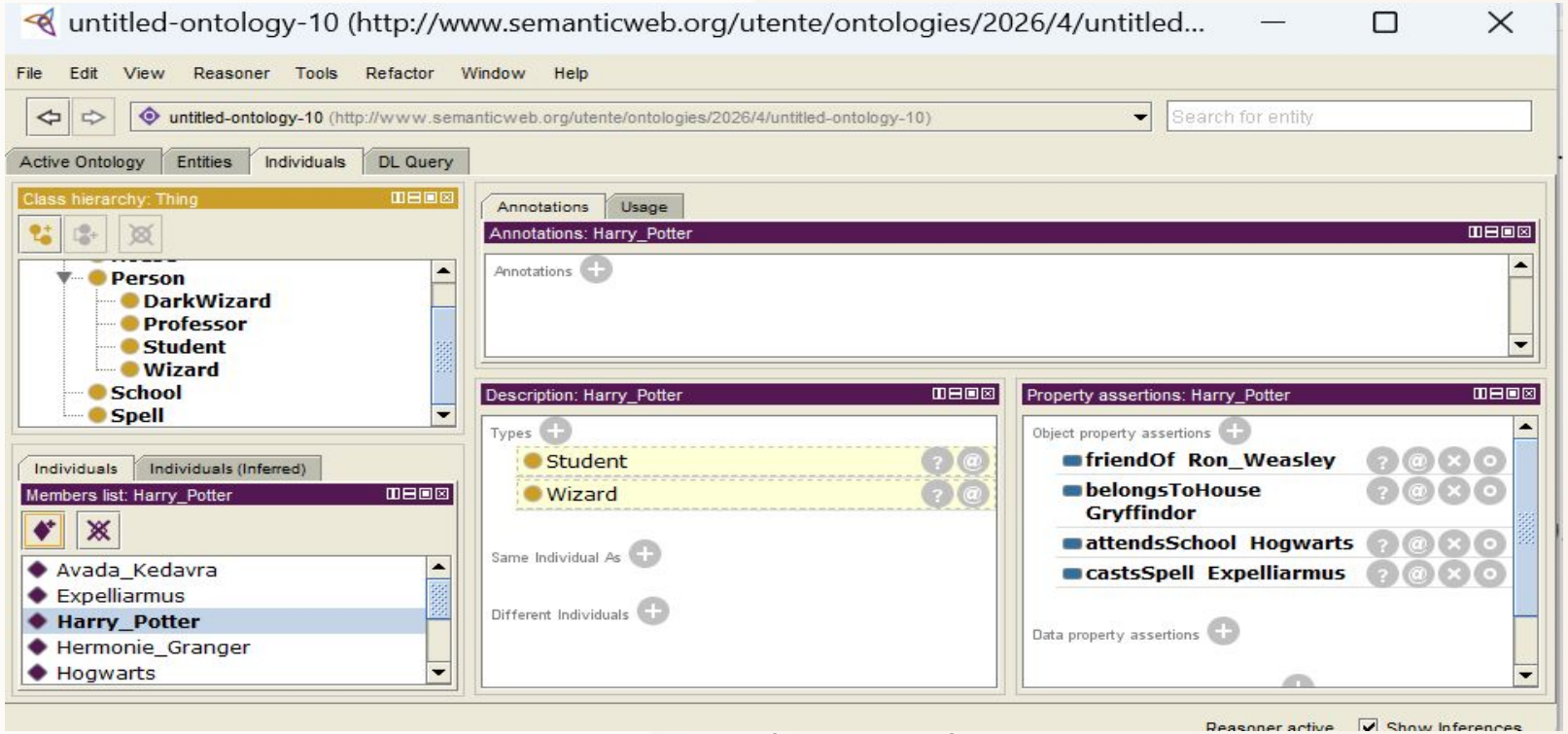
Note: *DarkWizard is kept as a sibling of Wizard (not a subclass) for modeling consistency; a future refinement could define DarkWizard ⊑ Wizard to inherit Wizard properties automatically.*

Object Properties & Reasoning

Example inference

- The HermiT reasoner was used to validate ontology consistency and infer implicit knowledge.

Property	Domain	Range
belongsToHouse	Student	House
friendOf	Student	Student
enemyOf	Wizard	Wizard
attendsSchool	Student	School
teaches	Professor	Spell
castsSpell	Wizard	Spell



*If:Harry **castsSpell** Expelliarmus and Domain(castsSpell)=Wizard then HermiT infers: Harry rdf:type **Wizard***

Key Modeling Choices & Justifications

Reasoning & Logic:

Symmetric Properties: friendOf and enemyOf were defined as symmetric.
Justification: Many relationships in fiction are bidirectional. The reasoner automatically infers both directions (e.g., if Harry is a friend of Ron, then Ron is a friend of Harry).

Disjoint Classes: We enforced mutual exclusivity with Professor ⊥ Student.
Justification: Prevents semantic contradictions and ensures that no individual can be logically classified as both a Professor and a Student simultaneously.

Additional disjoint axioms: House ⊥ Person and House ⊥ Spell.

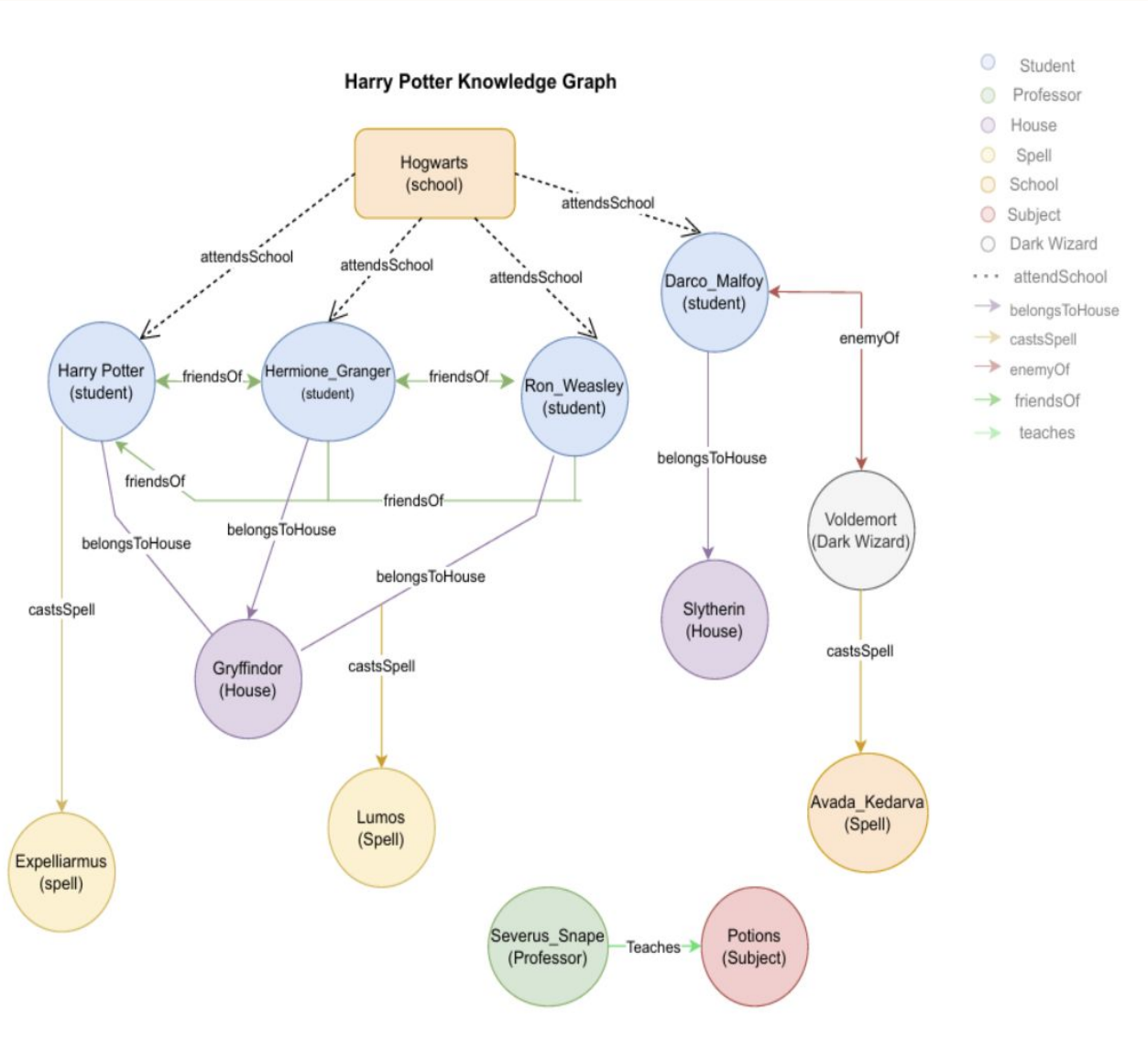
Integrity & Properties:

- **Existential Restriction**: Student ⊆ ∃ belongsToHouse.House.
  Justification: Guarantees that every Student belongs to at least one House, maintaining knowledge graph integrity and preventing incomplete data.
- **Inverse Properties**: teaches is defined as the inverse of taughtBy.
  Justification: Enables bidirectional querying, allowing the system to answer questions starting from either the Professor or the Spell.
- **Overall Approach**: These modeling choices follow GOLEM's flexible character modeling (Pianzola et al., 2025) and Scotti et al.'s (2025) emphasis on logical consistency and ontological rigor.

Manual Knowledge Graph Construction

Knowledge Graph Construction

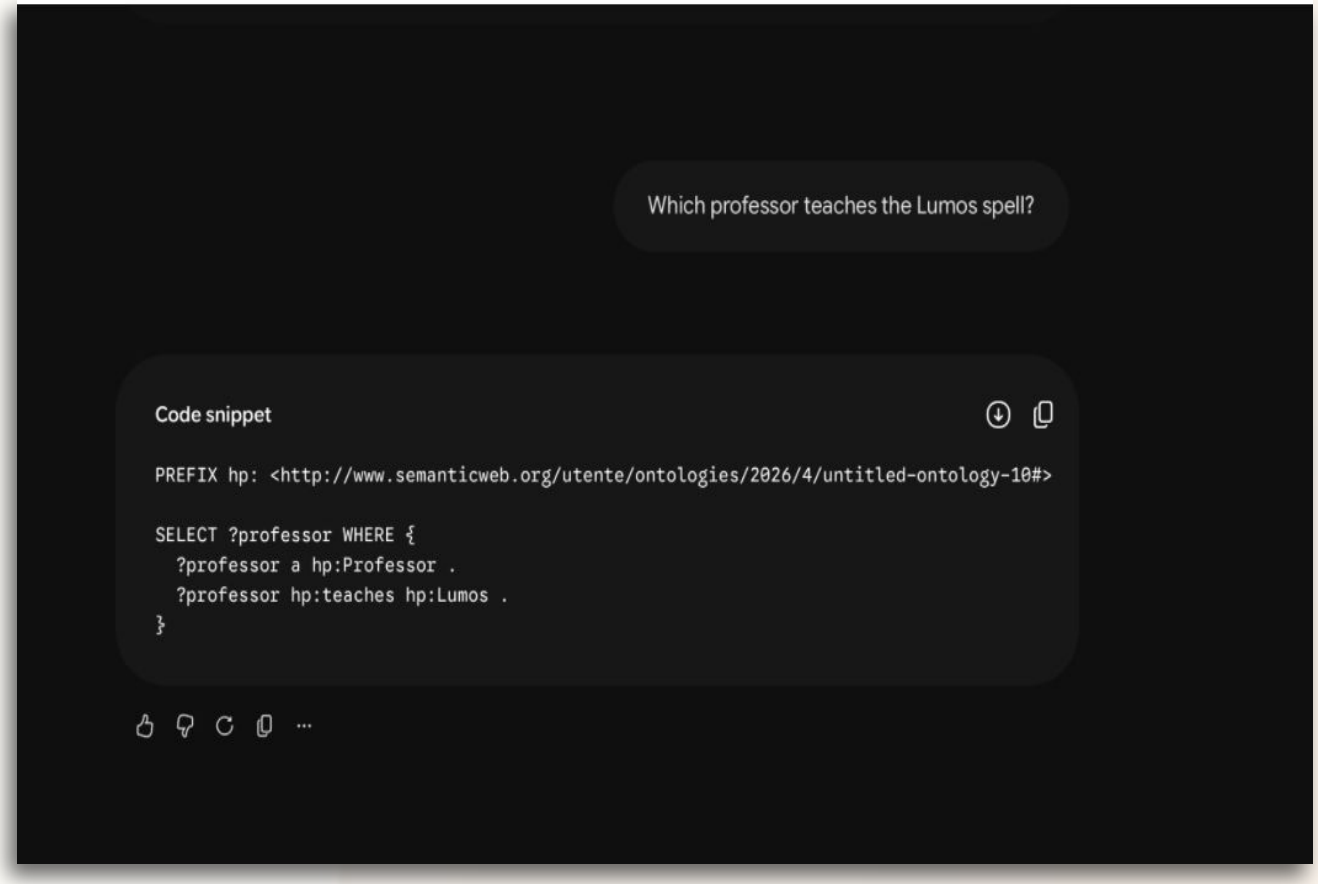
- **Domain Representation:** The graph captures key entities and relationships within the Harry Potter domain, including characters, houses, spells, subjects, and institutions.
- **Semantic Categories:** Node colors represent different ontology classes such as Students, Professors, Houses, Spells, and Dark Wizards.
- **Relationship Representation::** Edges correspond to ontology object properties, including belongsToHouse, friendOf, enemyOf, attendsSchool, teaches, and castsSpell.
- **Ontology Blueprint:** This graph served as the conceptual foundation for designing the OWL ontology in Protégé.



AI Prompting in Action

Performance Results

- **Syntax Generation:** Successfully translated complex natural language questions into valid, executable SPARQL code.
- **Query Capability:** Handled multi-hop reasoning, including *House affiliations*, *Spell-to-Professor mappings*, and *character social circles*.
- **Schema Mapping:** Effectively aligned the LLM's output with our custom OWL classes, properties, and constraints.
- **Technical Insight:** Optimized performance through *few-shot prompting*, guiding the model to accurately understand our specific domain logic.



KGQA Evaluation

Evaluation Methodology

To evaluate the Knowledge Graph Question Answering (KGQA) system, we created a set of 22 natural language competency questions.
For each question:

- A SPARQL query was generated via LLM.
- The query was executed in GraphDB.
- Returned results were compared with expected answers.

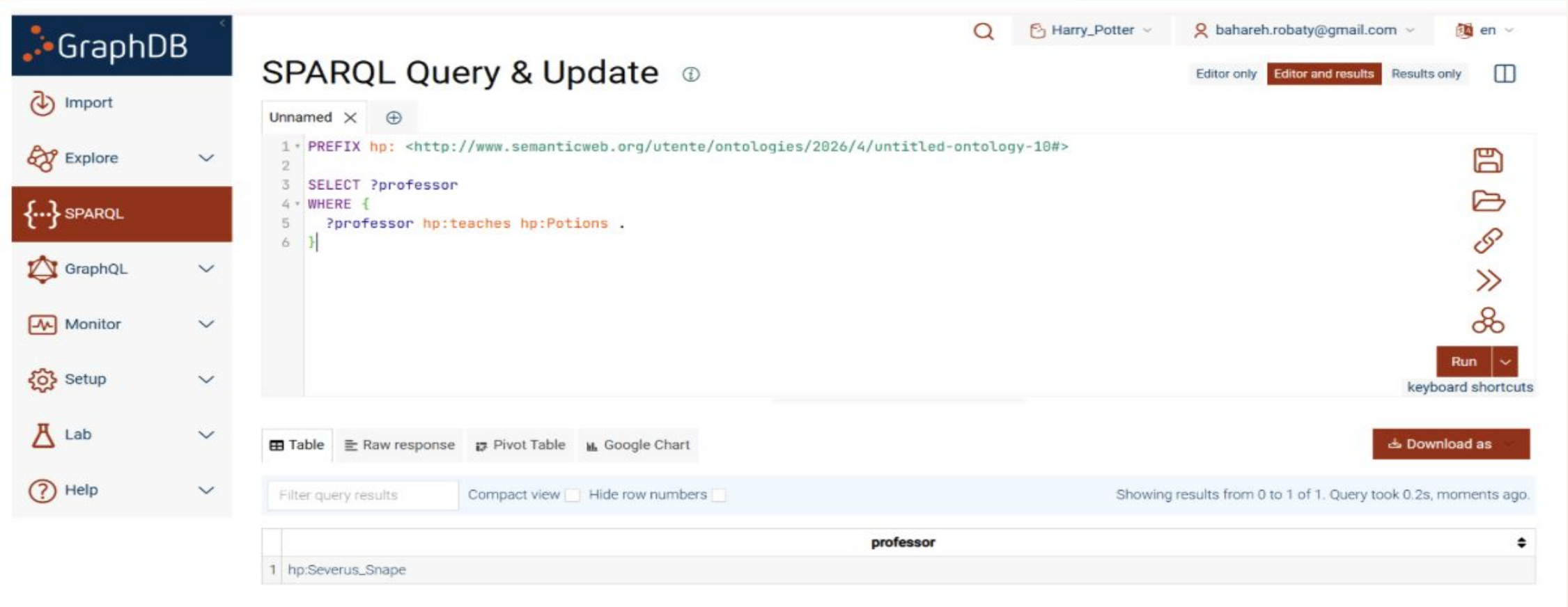
Metric	Value
Total Questions	22
Correct Answers	20
Failed Queries*	2
Consistency check	passed

*Result: The system successfully answered 20 out of 22 competency questions, **achieving a 90.9% accuracy rate**, and demonstrated effective integration between Natural Language input, SPARQL generation, and ontology querying.*
**Failure causes are detailed in Error Analysis (slide 14).*

SPARQL Queries

Query Example

"Which professor teaches Potions?"



SPARQL Queries

Question : Which students belong to Gryffindor?

SPARQL Query & Update

Unnamed ×

```
PREFIX hp: <http://www.semanticweb.org/utente/ontologies/2026/4/untitled-ontology-10#>

SELECT ?student
WHERE {
    ?student hp:belongsToHouse hp:Gryffindor .
}
```

Table | Raw response | Pivot Table | Google Chart

Filter query results | Compact view | Hide row numbers

	student
1	untitled-ontology-10:Harry_Potter
2	untitled-ontology-10:Hermione_Granger
3	untitled-ontology-10:Ron_Weasley

Question: Who is Draco Malfoy's enemy?

SPARQL Query & Update

Unnamed ×

```
PREFIX hp: <http://www.semanticweb.org/utente/ontologies/2026/4/untitled-ontology-10#>

SELECT ?enemy
WHERE {
    hp:Darco_Malfoy hp:enemyOf ?enemy .
}
```

Table | Raw response | Pivot Table | Google Chart

Filter query results | Compact view | Hide row numbers

	enemy
1	untitled-ontology-10:Voldemort

Error Analysis

Error 1 :

- **Question:** Who are all Wizards?
- **Cause:** Inferred class memberships generated by the Protégé reasoner were not materialized as explicit RDF triples in GraphDB.

Error 2 :

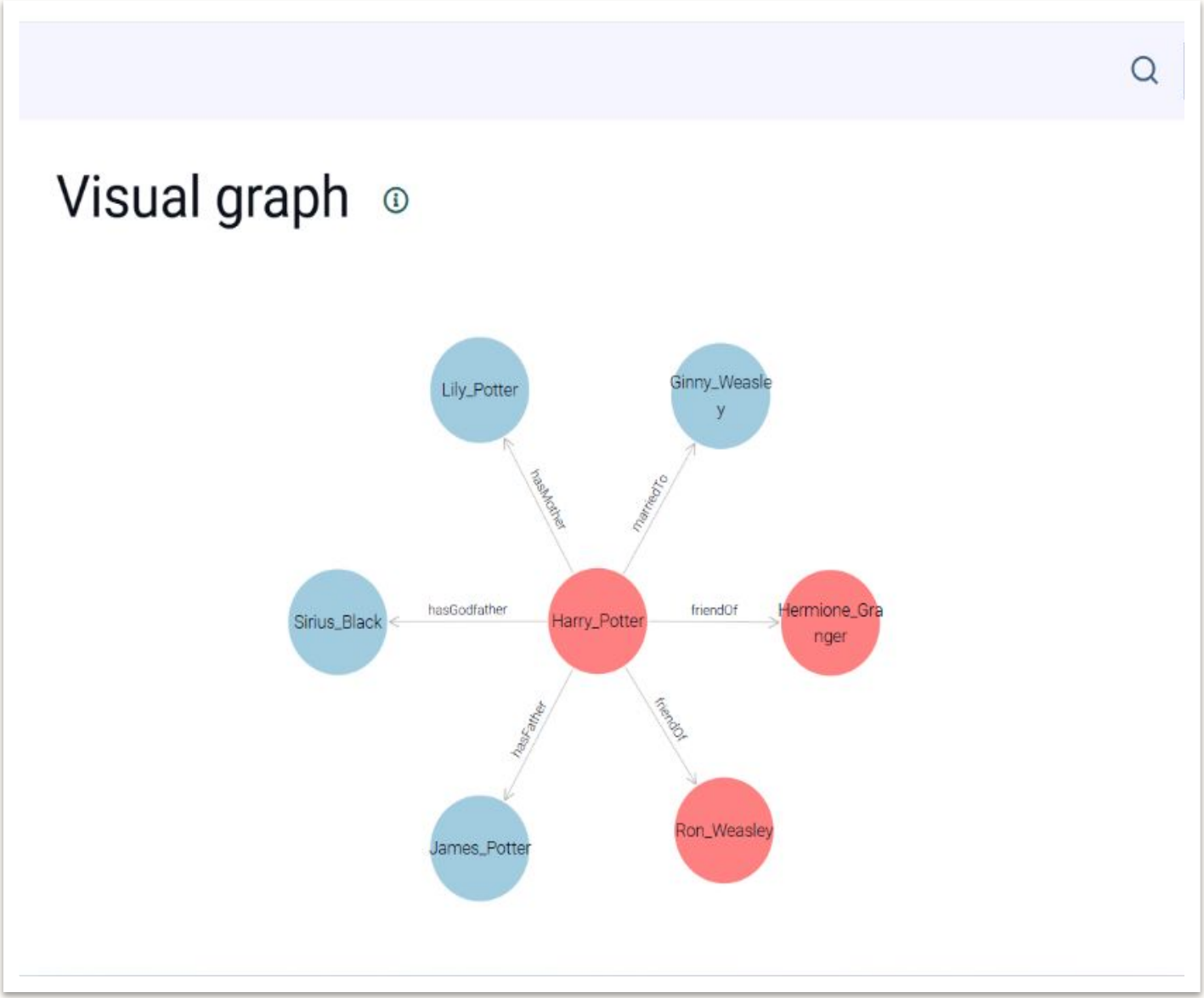
- **Question:** Which individuals cast some Spell?
- **Cause:** The query relied on OWL existential restrictions and inferred knowledge, while GraphDB only returned explicitly asserted relationships.

Lessons Learned:

- GraphDB primarily queries explicit RDF triples unless reasoning is enabled and configured.
- Some expected answers depended on inferred knowledge generated by the ontology reasoner.
- Query failures were caused by reasoning limitations, not ontology modeling errors.

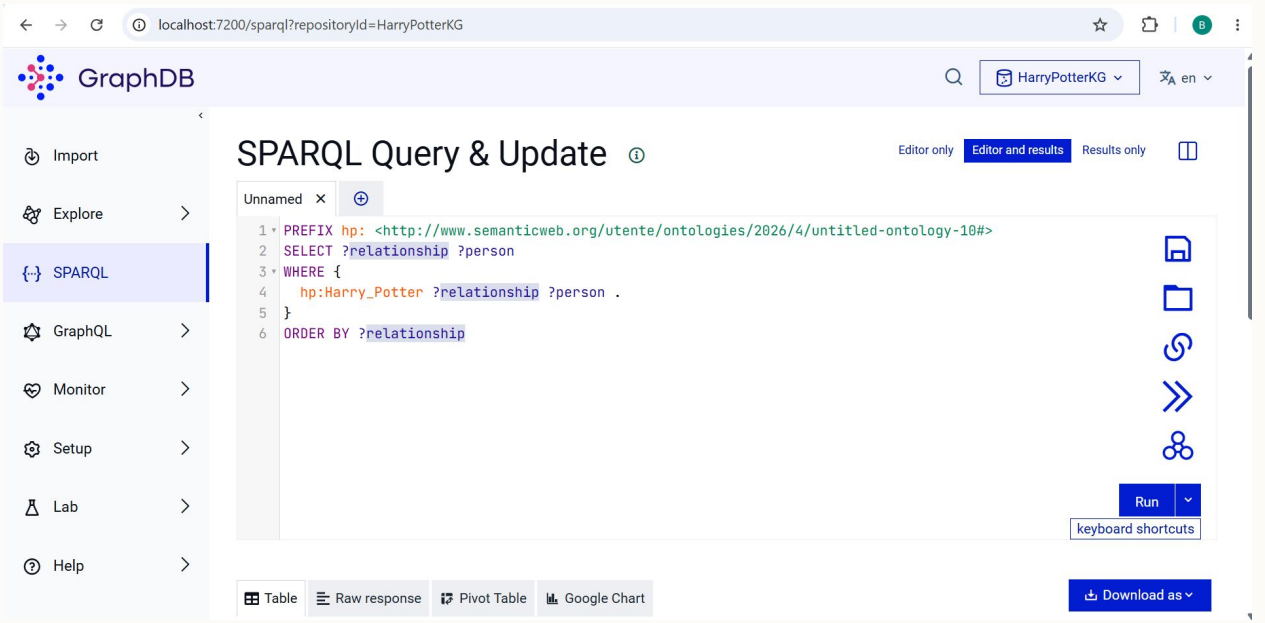
Knowledge Enrichment Using External Sources

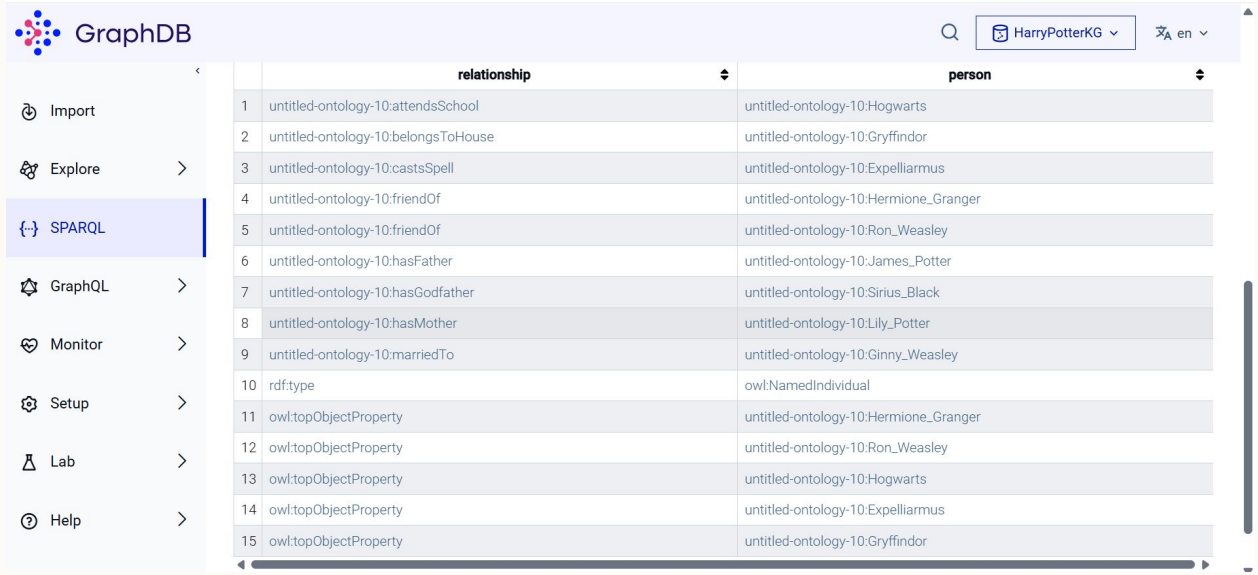
- Collected additional knowledge from Harry Potter Fandom Wiki
- Used Gemini (LLM) to extract RDF-style triples
- Converted the extracted data into RDF/Turtle format (harry_potter_import.ttl)
- Imported both the main ontology (.owl) and the new data into GraphDB
- The goal was to transform natural language knowledge into machine-readable semantic relationships.



SPARQL Validation

- Queried the enriched knowledge graph using SPARQL in GraphDB.
- Verified that newly imported RDF triples were correctly integrated into the ontology.
- Retrieved family relationships (e.g., *hasFather*, *hasMother*, *hasGodfather*).
- Validated social relationships such as *friendOf* and *marriedTo*.
- Confirmed consistency between manually modeled knowledge and LLM-extracted knowledge.



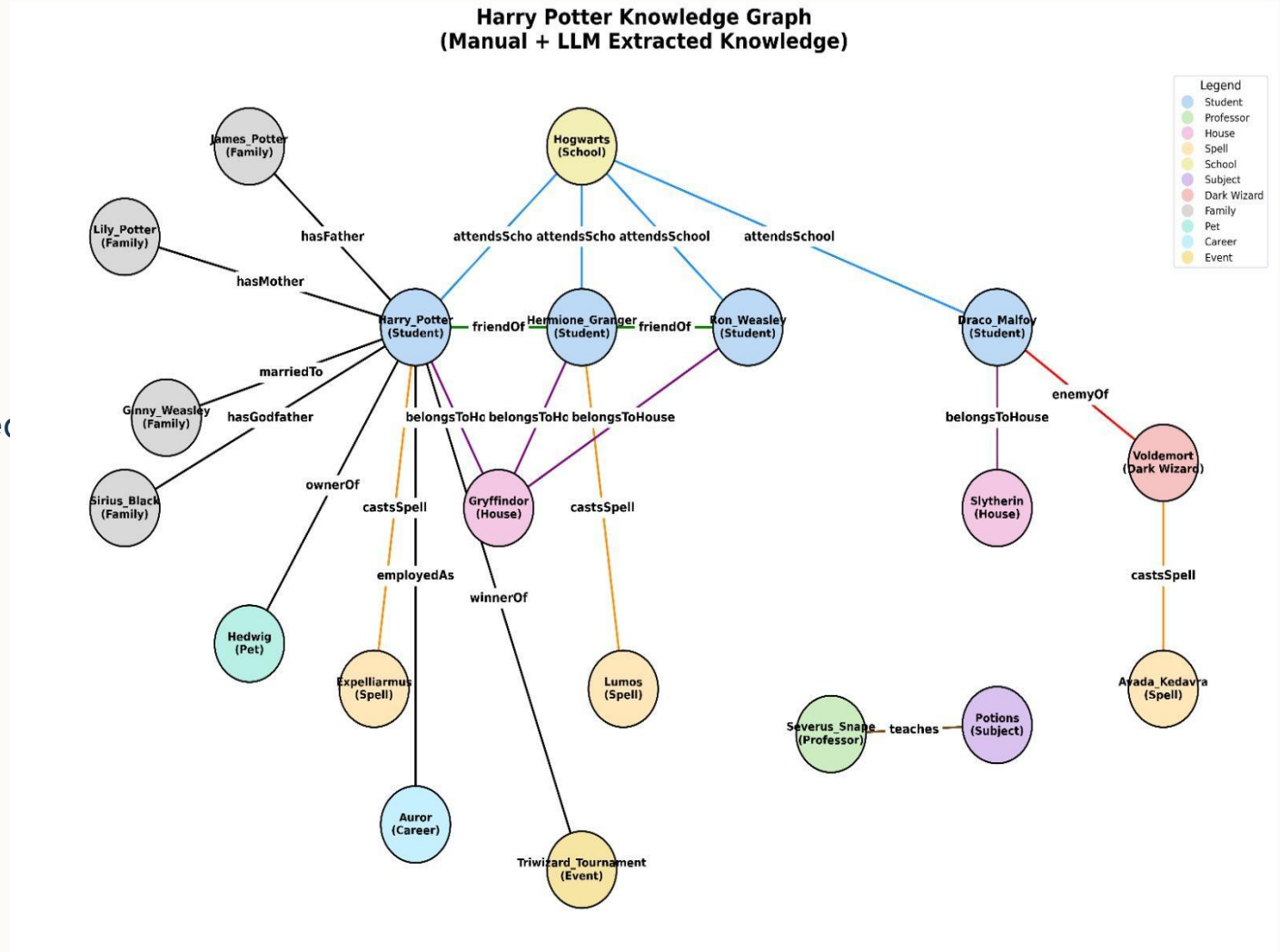


	relationship	person
1	untitled-ontology-10:attendsSchool	untitled-ontology-10:Hogwarts
2	untitled-ontology-10:belongsToHouse	untitled-ontology-10:Gryffindor
3	untitled-ontology-10:castsSpell	untitled-ontology-10:Expelliarmus
4	untitled-ontology-10:friendOf	untitled-ontology-10:Hermione_Granger
5	untitled-ontology-10:friendOf	untitled-ontology-10:Ron_Weasley
6	untitled-ontology-10:hasFather	untitled-ontology-10:James_Potter
7	untitled-ontology-10:hasGodfather	untitled-ontology-10:Sirius_Black
8	untitled-ontology-10:hasMother	untitled-ontology-10:Lily_Potter
9	untitled-ontology-10:marriedTo	untitled-ontology-10:Ginny_Weasley
10	rdf:type	owl:NamedIndividual
11	owl:topObjectProperty	untitled-ontology-10:Hermione_Granger
12	owl:topObjectProperty	untitled-ontology-10:Ron_Weasley
13	owl:topObjectProperty	untitled-ontology-10:Hogwarts
14	owl:topObjectProperty	untitled-ontology-10:Expelliarmus
15	owl:topObjectProperty	untitled-ontology-10:Gryffindor

Knowledge Graph Visualization by python

Knowledge Graph Construction

- Combined ontology data and LLM-extracted RDF triples.
- Exported all RDF/Turtle knowledge into a unifie graph structure.
- Generated a visual knowledge graph using Python.
- Nodes represent ontology entities and instances.
- Edges represent semantic relationships extracted from the ontology and RDF data.



Responsible AI Usage

- ChatGPT was used for project planning, ontology design discussions, report drafting, and presentation refinement.
- Gemini was used to transform unstructured textual information from external sources into RDF-style knowledge triples.
- AI tools assisted in knowledge extraction and representation, but ontology construction, class modeling, property definitions, and reasoning tasks were performed by the students.
- All generated RDF triples, SPARQL queries, and evaluation results were manually reviewed and validated before being incorporated into the final knowledge graph.

Limitations & Future Work

Limitations:

- Manual creation and population of individuals limits scalability.
- Some inferred knowledge generated in Protégé was not automatically materialized in GraphDB.
- Natural-language knowledge extraction depends on LLM output quality and may require manual validation.

Future Work:

- Automate knowledge extraction and RDF generation using LLMs and external knowledge sources.
- Integrate reasoning-aware pipelines to preserve inferred knowledge across platforms.
- Extend the ontology with events, timelines, locations, and additional character relationships.

Conclusion

The Harry Potter Knowledge Graph demonstrates how Semantic Web technologies can be used to model complex narrative domains.

Key Contributions: Designed an OWL ontology with meaningful classes, properties, restrictions, and disjointness axioms.
Applied automated reasoning using the HermiT reasoner to validate consistency and infer new knowledge.

Used LLMs to assist knowledge extraction and support semantic querying.

Final Takeaway: Combining ontology engineering, reasoning, and AI-assisted knowledge extraction provides a scalable framework for building enriched knowledge graphs.

"This project demonstrates how ontology engineering and modern AI techniques can work together to transform unstructured information into structured, queryable knowledge."

References

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Questions & Discussion

Thank you for your attention!

